

Stress-Testing Galaxy Formation In RAMSES

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Introduction

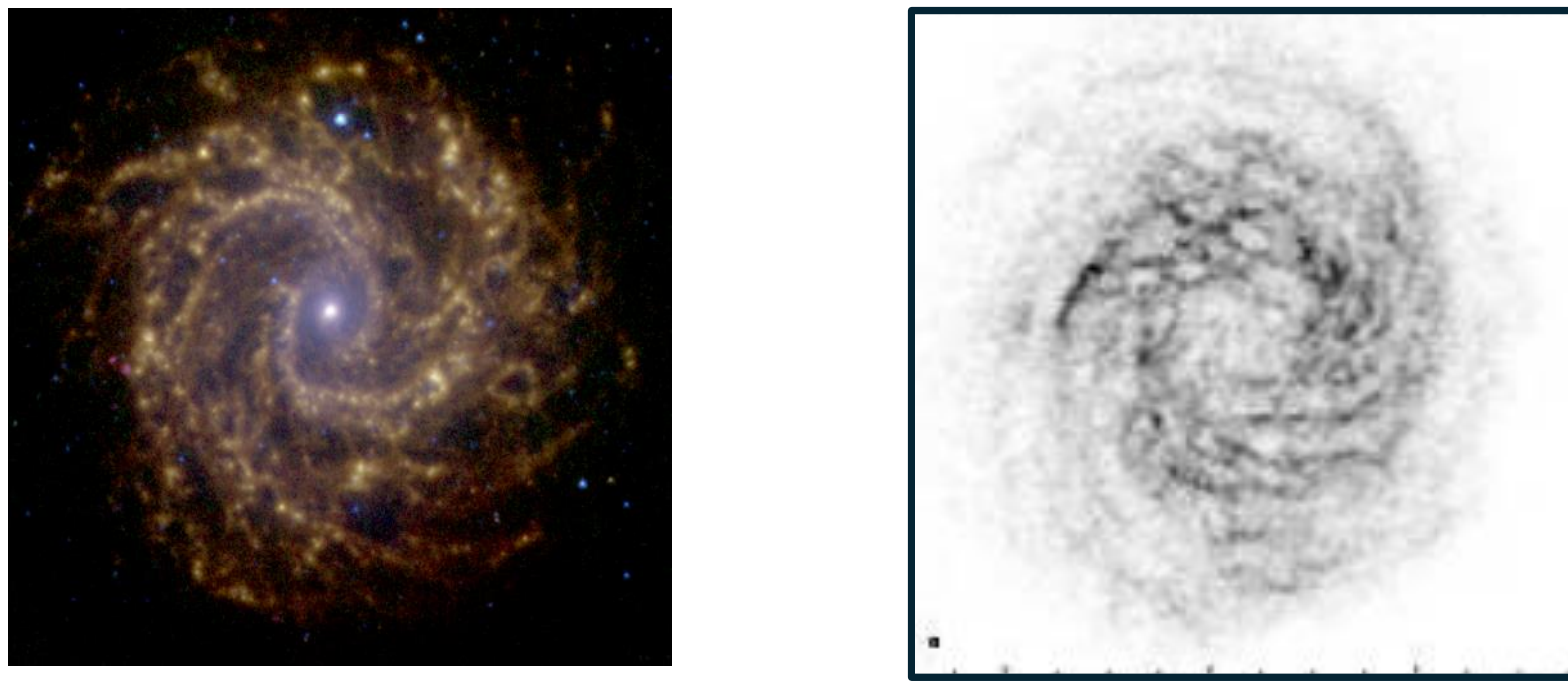


Fig 1. Left: False-colour infrared image of NGC 3184 [1], Right: Integrated HI (moment 0) map [5]

- Galaxies are self-gravitating objects comprised of stars, gas and dust, and dark matter.
- Galaxies undergo complex dynamics motions.
 - Gas turbulence ~ 10 km/s
 - Galactic rotation ~ 300 km/s
 - Galaxy-galaxy relative motion ~ 1000 km/s
- Hydrodynamics and gravity are Galilean invariant, but discretization in grid-based codes introduces numerical diffusion when a system moves relative to the grid, with the effect being more pronounced at supersonic speeds.

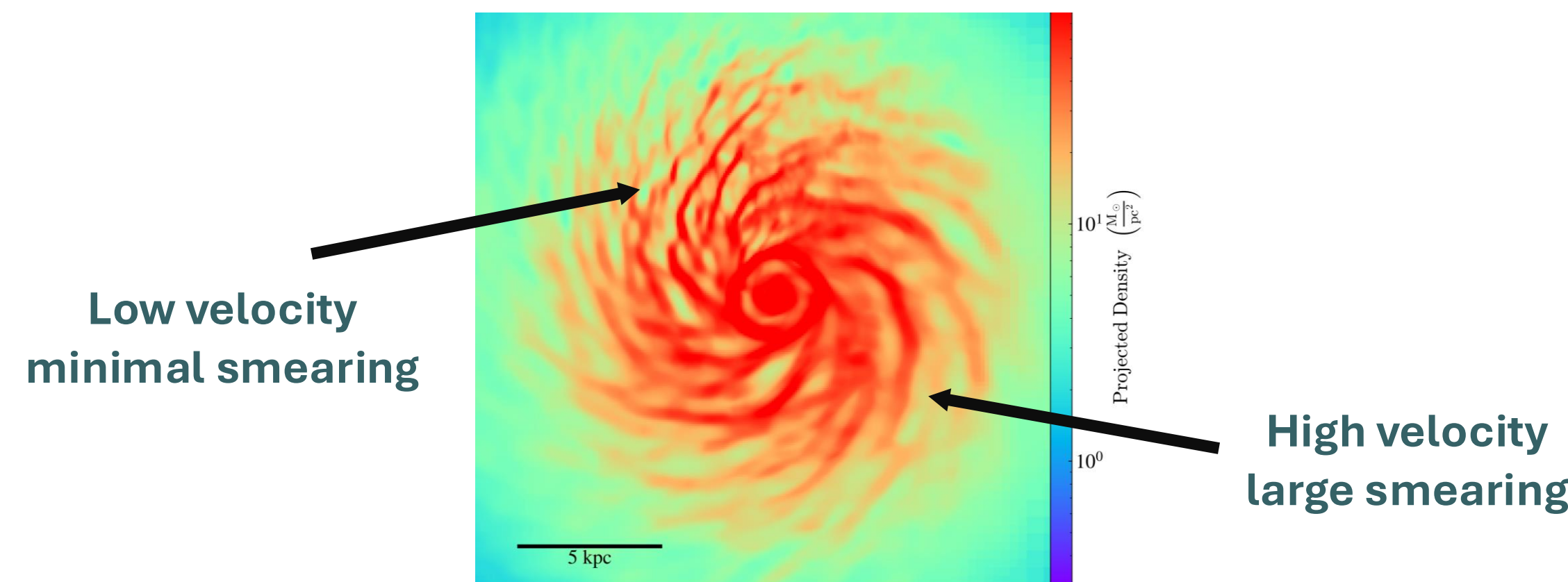


Fig 2. Galaxy simulation using AGORA initial conditions [2]. Regional smearing due to velocity gradient generated from translational and rotational velocity of magnitude $v = 200$ km/s

The purpose of this project is to assess the impact of deviations from Galilean invariance across different simulation setups.

Methods

- Modify and implement numerical astrophysics AMR code RAMSES [3].
- Simplify galaxy problem by examining simple perturbations in uniform boxes evolving under hydrodynamics and self gravity.
- Run simulations on HPC clusters and analyze data in Python using the yt package [4] to study density evolution, system stability, and velocity fields.

Results: Linear Tests

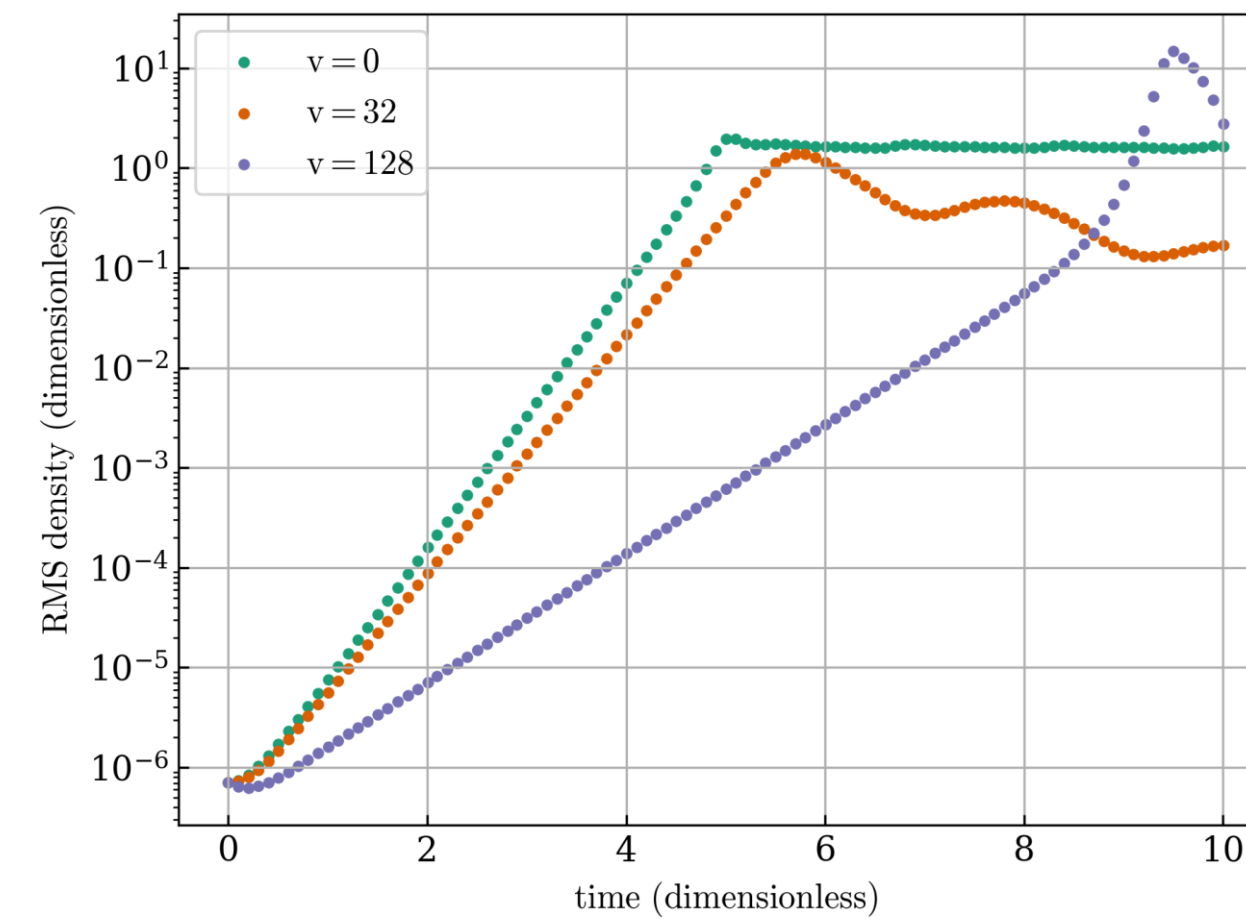


Fig 3. RMS density as a function of time for an unstable system with hydrodynamics and self-gravity, evolved with translational velocities $v = 0, 32$, and 128 along the positive x -direction.

- Used a small 10^{-6} amplitude wave to perturb a region with a uniform pressure and density.
- The system was initialized with constant translational velocities of magnitude $v = 2^n$ along a single axis in order to replicate different regions in a moving galaxy.
- Numerical diffusion is expected to scale as $D \sim \alpha(c_s + |v|)\Delta x$, where α is a dimensionless constant, Δx is the cell size, and $c_s + |v|$ represents the effective velocity.
- A hydrodynamics only test concluded that the diffusion followed the expected model and can be measured.
- When self gravity and hydrodynamics were tested, despite the increase in velocity, the linear perturbations continued to grow, and a collapse occurred in all cases.
- This indicated that the behaviour with gravity is more complex and a linear model is not sufficient to quantify the numerical diffusion that is observed in simulations.

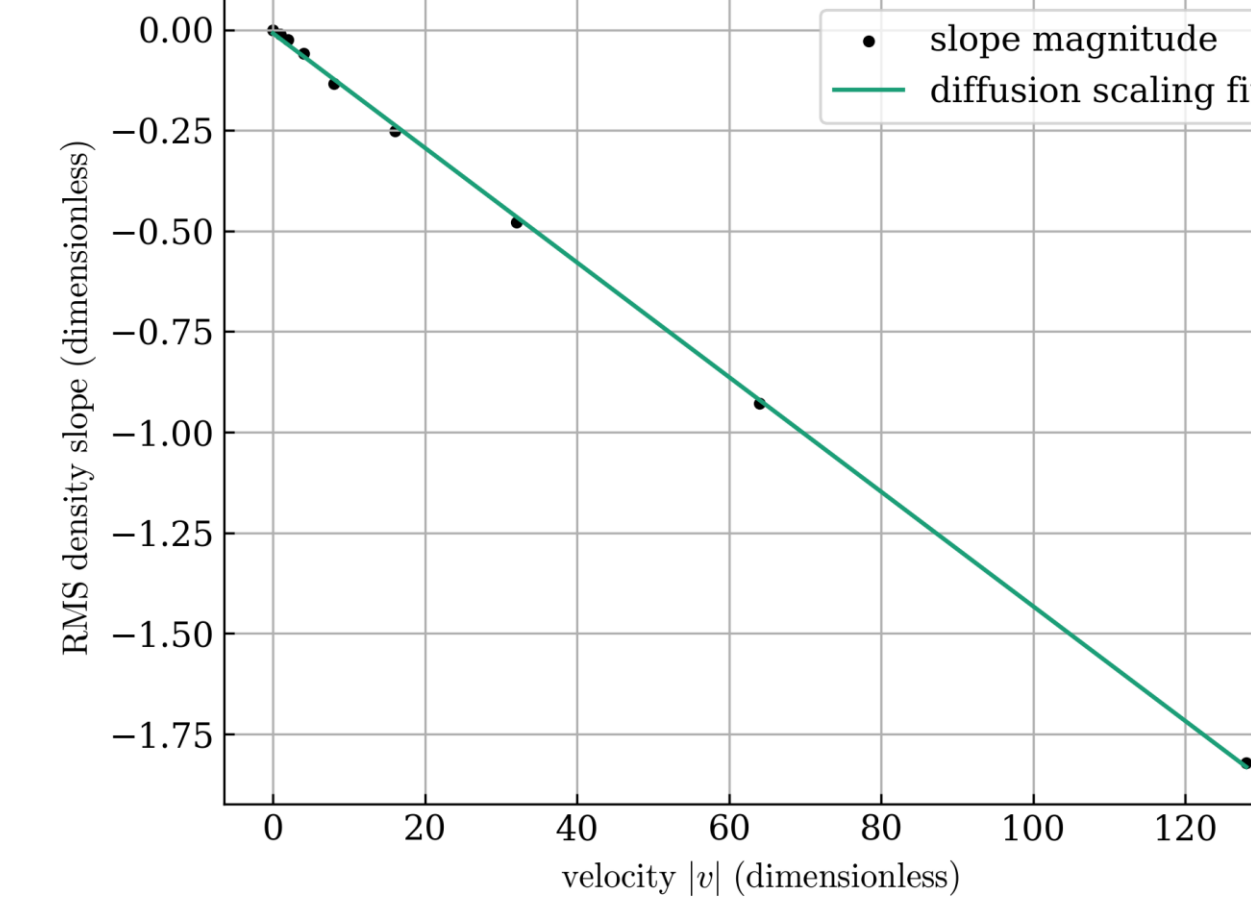


Fig 4. Slope of RMS density as a function of velocity for a stable system with hydrodynamics.

Results: Non-Linear Tests

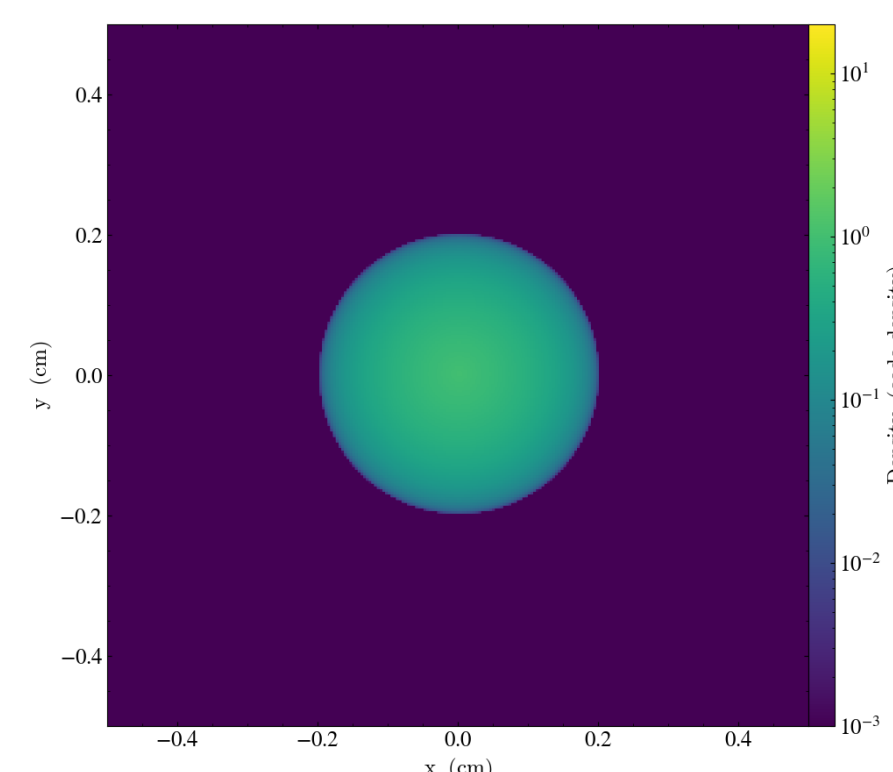


Fig 5. Density slice showing the initial state of a spherical blob with linearly decreasing density and pressure in an isothermal gas.

- Initialized a 3D box with a uniform background density and pressure consistent with an isothermal EOS.
- Spherical blob placed at the center with central density $\rho_c = 1.0$ and a density distribution that linearly decays with the radius.
- This setup reproduces non-linear gravitational collapse analogous to the Jeans instability, enabling the study of mode growth beyond the linear regime.
- The resulting behavior is complex and requires further analysis to determine the effects of velocity.

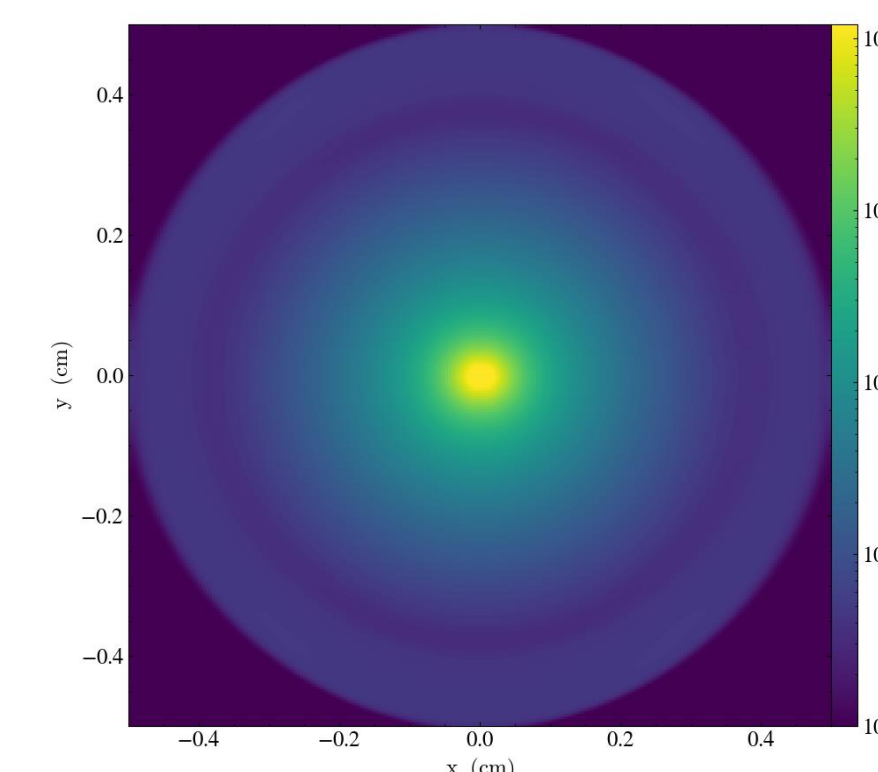


Fig 6. Density slice of the blob shown in Fig. 5 at a later time, illustrating central gravitational collapse and radial expansion of the structure.

Future Work

- Study the influence of velocity in the non-linear regime.
- Consider the influence of numerical and gravitational timescales when constructing simulations.
- Develop a model to quantify numerical diffusion using a non-linear framework.
- Define quality criteria that determines when numerical diffusion renders results unphysical

References

- [1] Boquien, M. (2006). *NGC 3184: Infrared image at 3.6, 5.8, and 8.0 μ m* [Image]. Wikimedia Commons. https://commons.wikimedia.org/wiki/File:NGC3184_3.6_5.8_8.0_microns_spitzer.png
- [2] Kim, J.-h., Abel, T., Agertz, O., et al. 2014, *The AGORA High-Resolution Galaxy Simulations Comparison Project, The Astrophysical Journal Supplement Series*, 210, 14, doi:10.1088/0067-0049/210/1/14
- [3] Teyssier, R. 2002, *Astronomy & Astrophysics*, 385, 337, doi:10.1051/0004-6361:20011817
- [4] Turk, M. J., Smith, B. D., Oishi, J. S., Skory, S., Skillman, S. W., Abel, T., & Norman, M. L. 2011, *ApJS*, 192, 9, doi:10.1088/0067-0049/192/1/9
- [5] Walter, F., Brinks, E., W J G de, B., Bigiel, F., Kennicutt, R. C., Jr., . . . Leroy, A. K. (2008). *THINGS: The HI nearby galaxy survey*. Ithaca: doi:<https://doi.org/10.1088/0004-6256/136/6/2563>